

Sahand Rahmdel

EDUCATION

M.Sc. Computer Science, Specialization: Artificial Intelligence and Data

RWTH Aachen University

Aachen, Germany
Apr. 2025 – Est. Mar. 2027

- *Relevant Coursework:* Machine Learning, Advanced ML, Data Science, Reinforcement Learning, Computer Vision, Business Process Intelligence, Actions & Planning in AI, Quantum Computing
- *Seminar Neuroinspired Computing – Reinforcement Learning* [\[Link\]](#): A paper on computational framework and biological foundations of Reinforcement Learning, Grade 1.0.

B.Sc. Computer Science including Coursework in Business Administration

RWTH Aachen University

Aachen, Germany
Nov. 2020 – Mar. 2025

- *Seminar Neuroinspired Computing – Attention Mechanism and its Applications* [\[Link\]](#): A paper on biological vs. computational Attention Mechanism; presented a deep-dive on the Transformer, Grade 1.3.

WORK EXPERIENCE

AI Engineer (Working Student)

Mercedes-Benz AG

Nov. 2025 – Present
Sindelfingen, Germany - Remote

- Sole engineer and technical lead contact of NEXA (NX Engineering eXpert Agent), an enterprise AI agentic companion; presented live architecture to 800+ attendees (500 in-person) at company town hall.
- Architected NEXA's multi-agent LangGraph pipeline with local RAG (semantic and lexical search tools with reranking) and custom MCP services to automate code generation across a 70k+ LOC enterprise C# repository.
- Engineered core modules for TQA, a Siemens NX CAD conformance checker; optimized unused-object analyzer to improve speed by 47%.

Research Assistant – HPC for NLP

Jülich Supercomputing Centre (JSC), Forschungszentrum Jülich

Apr. 2024 – Dec. 2024
Jülich, Germany

- Built HPC evaluation pipelines to construct a multilingual embedding proximity map across 24 languages for Teuken-7B (Germany's 1st LLM), optimizing cross-lingual transfer learning and data selection.

PROJECTS

HackerRank LLM Orchestration Hackathon | Top 4% Worldwide [\[Link\]](#)

Jun. 2026

- Built a multi-modal VLM claim reviewer with deterministic guardrails for business logic, ranking 2nd in Germany.

Software Project Lab – Artifaix (AI-supported Information Foraging) [\[Link\]](#)

Oct. 2025 – Apr. 2026

Research Group Software Construction (SWC), RWTH Aachen

Aachen, Germany

- Led a 4-person team to design a modular RAG-based microservice architecture for automated knowledge extraction.
- Implemented LLM reasoning and MapReduce processing for papers exceeding standard model context windows.

Bachelor's Thesis – Exploring Linguistic Proximity [\[Link\]](#)

Jun. 2024 – Dec. 2024

Visual Computing Institute, RWTH Aachen & Forschungszentrum Jülich & WestAI

Aachen, Germany

- Analyzed 6.1 TB of multilingual text data from 24 languages, applied t-SNE to map high-dimensional clusters discovering that raw embeddings cluster by topic, and mean embeddings cluster by linguistic family.

Software Project Lab – Network Engineering Practical Internet eXperience (PIX)

Oct. 2022 – Apr. 2023

Chair of Communication and Distributed Systems (ComSys), RWTH Aachen

Aachen, Germany

- Configured OSPF, BGP, and RPKI protocols in a simulated network of 28 virtual routers.

System Programming Lab – Embedded Systems Programming

Oct. 2021 – Apr. 2022

Chair of Embedded Software, RWTH Aachen

Aachen, Germany

- Developed low-level C firmware for ATmega644 embedded system, managing memory and peripherals (GPIO, Timers).

TECHNICAL SKILLS

Programming: Python, C# (.NET), Java, C/C++ , Bash, TypeScript, HTML/CSS

Machine Learning & NLP: PyTorch, Hugging Face (Transformers, Datasets), Sentence Transformers, scikit-learn, Vector Embeddings, RAG, LangChain, LangGraph, ChromaDB, MCP

Data & Process Mining: pandas, NumPy, Data Visualization, ProM, Celonis

Backend & Systems: FastAPI, Microservices, LINQ, async/await, Linux, Docker, SLURM, MapReduce, Vector Databases

Databases & Infrastructure: MySQL, SQL, Git, Conda/venv/uv

Frontend & Collaboration: React, Vite, Tailwind CSS, shadcn/ui, Jira, Confluence

Engineering Tools: ~~ETX~~, RapidMiner, Siemens NX, NX Open API, Visual Studio

LANGUAGES

German: DSH-3 (C2 / Fluent)

English: Fluent

Persian: Native

Azerbaijani: Native

Turkish: Fluent